

Exact Sparse Incidence Decomposition for Provenance Completion Programs: Componentwise Primal–Dual Certificates and Verified Carrier Census Protocols

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Abstract

TPC–76 reduces every fixed-row-pair provenance completion problem to

$$\Gamma(b, C) = \min_{\theta \in \mathbb{C}^J} \|b + C\theta\|_1, \quad b = A_D F^0, \quad C = A_D B_L.$$

The formula is exact, but it does not say which parts of a literal carrier interact or how a finite optimum can be independently checked. This paper closes that sparse finite-dimensional layer.

We first separate zero columns, immutable rows, and active coordinates. Let $\mathcal{G}_C$ be the bipartite graph on active analysis rows and active filter variables, with $i \sim j$ exactly when the fully assembled coefficient C_{ij} is nonzero. If (I_s, J_s) are its connected components, then

$$\Gamma(b, C) = \|b_{\text{fix}}\|_1 + \sum_s \gamma_s, \quad \gamma_s = \min_{\theta_s} \|b_s + C_s \theta_s\|_1.$$

This is an exact factorization, not a sparsity heuristic. Its dual factors over the same components:

$$\gamma_s = \max_{\substack{\|z_s\|_\infty \leq 1 \\ C_s^* z_s = 0}} \text{Re}\langle z_s, b_s \rangle.$$

For every trial fill θ_s and exact dual-feasible charge z_s , the quantities

$$U_s = \|b_s + C_s \theta_s\|_1, \quad L_s = \text{Re}\langle z_s, b_s \rangle$$

satisfy $L_s \leq \gamma_s \leq U_s$, and their gaps add exactly. Equality is equivalent to phase saturation of the residual. Thus each reported finite value can carry a matched fill/charge certificate.

We also compare $\mathcal{G}_C$ with the ambient filter–site–analysis tripartite graph. The ambient graph gives a safe coarse block decomposition, whereas exact cancellations in $C = A_D B_L$ may produce additional splitting. Thresholded floating support is not admissible for this purpose.

The specialization preserves the complete literal fixed- h_0 carrier and its original global raw masses. No archived literal coefficient census exists in the preceding papers, so no numerical physical census is invented here. The results are L0 component theorems and an L1 certificate protocol, not an L2 saving, Perron decay, fixed-2, prime-pair, or twin-prime conclusion.

Keywords: sparse incidence; bipartite graph; convex duality; analytic completion; component factorization; exact certificate; fixed shift.

Convention. Every physical specialization retains one fixed nonzero shift h_0 , literal coefficients, complete native output keys, actual nonzero support, and global raw atomic normalization. Exact zero means algebraic zero, never a numerical threshold.

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1 Introduction

The completion route has reached a finite but nontrivial optimization problem. TPC-75 [2] proves strong duality for arbitrary nonprimitive completions. TPC-76 [3] restricts the free values to literal content-flow directions and obtains

$$\gamma_D^{\text{prov}}(\mathbf{L}) = \min_{\theta} \|A_D(F^0 + B_{\mathbf{L}}\theta)\|_1.$$

Set

$$b = A_DF^0, \quad C = A_DB_{\mathbf{L}}.$$

The remaining finite question is the distance of $-b$ from $\text{Ran } C$ in complex ℓ^1 .

The matrix C , however, is highly structured. A filter variable first copies literal q -slice amplitudes onto permitted nonprimitive sites; the divisor-variation operator then sends those sites to low plaquettes and high-tail coordinates. Many formal variables vanish, many analysis rows are immutable, and some groups cannot interact. Ignoring this structure obscures both computation and proof.

1.1 Why the product $C = AB$ is the correct graph

It is tempting to draw a graph from the support of $B_{\mathbf{L}}$ and the stencil of A_D . That graph is useful, but it is not necessarily sharp. For fixed analysis row i and filter column j ,

$$C_{ij} = \sum_k (A_D)_{ik} (B_{\mathbf{L}})_{kj}.$$

Distinct site contributions can cancel exactly. The maximal support-based decomposition therefore uses the nonzero entries of the assembled C , after all literal sums and signs are fixed. The tripartite graph will be proved to give a safe coarser partition.

1.2 Main advance

The paper proves four exact statements.

- (i) immutable coordinates and cost-null variables have different deletion rules;
- (ii) the actual incidence graph of C block-diagonalizes the affine ℓ^1 distance exactly;
- (iii) the same components factor the strong dual, including the fixed rows; and
- (iv) matched primal and dual data give a locally and globally checkable gap certificate.

These statements turn a monolithic solver output into a sum of independent mathematical claims.

1.3 What is deliberately absent

The previous TPC papers define the literal carrier symbolically but do not archive a table of all active $L_{mn,q}$ coefficients on a declared growing block. Consequently this paper proves the exact census protocol but does not fabricate a numerical census. Even a completed finite census would remain L1: asymptotic normalized component mass, row sums, Perron decay, shared-rule uniformity, and physical reassembly would still be open.

1.4 Organization

Section 2 fixes the realified complex normal form. Section 3 proves exact deletion rules. Section 4 gives primal component factorization. Section 5 gives the dual and gap certificates. Section 6 compares actual and ambient graphs. Section 7 states the literal fixed-shift protocol. Section 8 records non-substitution examples, and section 9 gives the claim ledger.

2 Finite realified normal form

2.1 Spaces and pairing

Let I and J be finite index sets. Regard $\mathbb{C}^I, \mathbb{C}^J$ as real vector spaces and use

$$[x, y]_{\mathbb{R}} = \operatorname{Re}\langle x, y \rangle = \operatorname{Re} \sum_i \bar{x}_i y_i.$$

The adjoint of a real-linear map C is characterized by

$$[C\theta, z]_{\mathbb{R}} = [\theta, C^*z]_{\mathbb{R}}.$$

Complex-linear provenance filters are the main case, while the real-linear convention also covers a shared filter constrained by conjugation.

For a real-linear map, C_{ij} below denotes the real-linear coordinate block from the j -th copy of $\mathbb{C}$ to the i -th copy, not necessarily multiplication by one complex scalar. Thus $C_{ij} \neq 0$ means that this block map is nonzero. In the complex-linear pair-dependent program, it is the usual scalar matrix entry.

Definition 2.1 (Affine completion value). For $b \in \mathbb{C}^I$ and a real-linear map $C : \mathbb{C}^J \rightarrow \mathbb{C}^I$, define

$$\Gamma(b, C) = \min_{\theta \in \mathbb{C}^J} \|b + C\theta\|_1.$$

The minimum exists. Indeed, $b + \operatorname{Ran} C$ is a closed affine subspace in finite dimension, and a nearest point to the origin exists. The parameter θ need not be unique.

2.2 Global dual

Finite-dimensional distance-to-a-subspace duality [1, Section 15] gives

$$\Gamma(b, C) = \max_{\substack{\|z\|_{\infty} \leq 1 \\ C^*z=0}} [z, b]_{\mathbb{R}}. \quad (1)$$

Both extrema are attained. The coordinatewise unit polydisc, rather than a real cube, is the dual ball of complex ℓ^1 .

2.3 Exact support

All graph statements use

$$\operatorname{supp} C = \{(i, j) \in I \times J : C_{ij} \neq 0\}$$

after exact assembly. If entries are validated intervals, an edge may be deleted only when its interval is the singleton $\{0\}$. A small floating magnitude is not an exact zero.

Remark 2.2 (Canonical coordinates). Changing bases in parameter space can alter the support graph while preserving $\operatorname{Ran} C$. The graph factorization below concerns the declared literal filter coordinates. It is maximal among decompositions read directly from that coordinate support, not among all possible algebraic decompositions of the quotient.

3 Exact deletion rules

Define

$$\begin{aligned} I_{\text{fix}} &= \{i \in I : C_{i,*} = 0\}, & I_{\text{act}} &= I \setminus I_{\text{fix}}, \\ J_0 &= \{j \in J : C_{*,j} = 0\}, & J_{\text{act}} &= J \setminus J_0. \end{aligned}$$

Definition 3.1 (Closed coordinate block). A pair $S = (I_S, J_S)$, with $I_S \subseteq I$ and $J_S \subseteq J$, is closed if

$$C[I_S, J \setminus J_S] = 0, \quad C[I \setminus I_S, J_S] = 0.$$

Thus its variables affect no outside rows and no outside variables affect its rows.

Theorem 3.2 (Sparse deletion normal form). *The following reductions are exact.*

- (a) *Every column in J_0 may be fixed canonically to zero without changing $\Gamma(b, C)$.*
- (b) *The rows in I_{fix} contribute the immutable amount*

$$\|b_{\text{fix}}\|_1 = \sum_{i \in I_{\text{fix}}} |b_i|.$$

- (c) *A fixed row with $b_i = 0$ may be removed from the reduced optimization data; zero columns are removed independently as in (a).*
- (d) *A closed coordinate block $S = (I_S, J_S)$ can be deleted as zero-cost if an exact vector θ_S satisfying $C_S \theta_S = -b_S$ is supplied. The assertion cannot be made from dimension count or solver failure alone.*

Proof. Zero columns do not occur in $C\theta$. Fixed rows equal b_i for every θ , so their absolute values split from the objective. The zero fixed coordinates contribute nothing. For a closed block the objective and parameter space split from the complementary coordinates. Finally, $\min_{\theta} \|b_S + C_S \theta\|_1 = 0$ if and only if $-b_S \in \text{Ran } C_S$, which is equivalent to the displayed exact equation. $\square$

Remark 3.3 (Cost-null is not provenance-null). A column can vanish in $C = A_D B_L$ even when the underlying column of B_L is nonzero. It is invisible to the declared cost, but it remains part of the content-flow manifest if an analytic fill is reconstructed. Deletion from optimization is not deletion from provenance.

Corollary 3.4 (Range and obstruction alternatives). *For any closed active block S , exactly one of the following holds:*

- (i) *$-b_S \in \text{Ran } C_S$, and the block value is zero; or*
- (ii) *there exists $z_S \in \text{Ker } C_S^*$ with $[z_S, b_S]_{\mathbb{R}} > 0$, and the block value is positive after normalizing $\|z_S\|_{\infty} \leq 1$.*

Proof. If $-b_S \notin \text{Ran } C_S$, finite-dimensional separation supplies a real functional annihilating $\text{Ran } C_S$ but not b_S . Change its sign and normalize. The converse follows by pairing with $b_S + C_S \theta_S = 0$. $\square$

Remark 3.5 (The offset must be retained). Building a graph only from C does not permit one to discard b . Rows with no incident column are immutable costs, and the values b_s on active components determine whether those components can be cancelled.

4 Actual-incidence component factorization

Definition 4.1 (Actual incidence graph). Let $\mathcal{G}_C$ be the bipartite graph with vertex set

$$I_{\text{act}} \sqcup J_{\text{act}}$$

and edge $i \sim j$ if and only if $C_{ij} \neq 0$. Write its connected components as

$$(I_s, J_s), \quad 1 \leq s \leq r,$$

and put

$$b_s = b|_{I_s}, \quad C_s = C[I_s, J_s].$$

No component consists of only one side: active rows and columns have at least one incident edge.

Theorem 4.2 (Exact primal component factorization). *One has*

$$\Gamma(b, C) = \|b_{\text{fix}}\|_1 + \sum_{s=1}^r \gamma_s, \quad \gamma_s = \min_{\theta_s \in \mathbb{C}^{J_s}} \|b_s + C_s \theta_s\|_1. \quad (2)$$

Every component minimum is attained. A global minimizer is exactly a tuple of component minimizers, together with arbitrary zero-column coordinates.

Proof. Permute active rows and columns by components. If $s \neq t$, then $C[I_s, J_t] = 0$, otherwise an edge would join the two components. Hence

$$C|_{I_{\text{act}} \times J_{\text{act}}} = \bigoplus_{s=1}^r C_s.$$

The parameter space is the Cartesian product of the component spaces, and the ℓ^1 norm is the sum over disjoint row sets. Fixed rows have already split off by [theorem 3.2](#). Therefore minimization separates into the displayed sum. Finite-dimensional attainment holds componentwise. $\square$

Corollary 4.3 (Exact component census identities). *The active counts satisfy*

$$|I_{\text{act}}| = \sum_s |I_s|, \quad |J_{\text{act}}| = \sum_s |J_s|, \quad \#\text{supp } C = \sum_s \#\text{supp } C_s.$$

The cost identity (2) is independent of the component ordering.

Remark 4.4 (Support maximality). $\mathcal{G}_C$ gives the finest decomposition obtained by permuting the declared coordinate rows and columns until C is block diagonal. Further reduction may be possible after a non-coordinate change of basis or a quotient by $\text{Ker } C$, but that is algebraic compression, not another connected-component statement.

4.1 Hermitian reversal

Proposition 4.5 (Component transport under reversal). *Suppose a reversed row pair has*

$$b_{nm} = \overline{Tb_{mn}}, \quad C_{nm} = \overline{TC_{mn}S},$$

where T, S are coordinate permutations induced by transposition and content-label reversal. Then $\mathcal{G}_{C_{nm}}$ is isomorphic to $\mathcal{G}_{C_{mn}}$, with the same component sizes and values.

Proof. Permutation and conjugation preserve exact zeros and the ℓ^1 -distance. A primal minimizer and dual charge transfer by the same maps. $\square$

5 Dual factorization and matched certificates

Theorem 5.1 (Exact dual component factorization). *For each active component,*

$$\boxed{\gamma_s = \max_{\substack{\|z_s\|_\infty \leq 1 \\ C_s^* z_s = 0}} [z_s, b_s]_{\mathbb{R}}.} \quad (3)$$

All maxima are attained. The global dual feasible set is the product of the component annihilators and the free unit discs on fixed rows. An optimal fixed-row charge is

$$(z_{\text{fix}})_i = \begin{cases} b_i/|b_i|, & b_i \neq 0, \\ 0, & b_i = 0. \end{cases}$$

Proof. Apply (1) to every C_s . Equivalently, after the component permutations,

$$C^* z = 0 \iff C_s^* z_s = 0 \quad (1 \leq s \leq r),$$

while fixed-row coordinates do not occur in $C^* z$. Their support function is $\sum_{i \in I_{\text{fix}}} |b_i|$, attained by the displayed phase. Objectives and constraints then factor. $\square$

Theorem 5.2 (Matched fill-charge certificate). *For arbitrary $\theta_s \in \mathbb{C}^{J_s}$ and exact dual-feasible $z_s \in \mathbb{C}^{I_s}$, define*

$$y_s = b_s + C_s \theta_s, \quad U_s = \|y_s\|_1, \quad L_s = [z_s, b_s]_{\mathbb{R}}.$$

Then

$$L_s \leq \gamma_s \leq U_s, \quad U_s - L_s = \|y_s\|_1 - [z_s, y_s]_{\mathbb{R}} \geq 0. \quad (4)$$

Globally,

$$\|b_{\text{fix}}\|_1 + \sum_s L_s \leq \Gamma(b, C) \leq \|b_{\text{fix}}\|_1 + \sum_s U_s, \quad (5)$$

and the global gap is exactly $\sum_s (U_s - L_s)$.

Proof. Weak duality gives the first inequalities. Since $C_s^* z_s = 0$,

$$[z_s, b_s]_{\mathbb{R}} = [z_s, b_s + C_s \theta_s]_{\mathbb{R}} = [z_s, y_s]_{\mathbb{R}}.$$

The coordinatewise bound $|z_{s,i}| \leq 1$ gives $[z_s, y_s]_{\mathbb{R}} \leq \|y_s\|_1$. Sum the component inequalities and use [theorem 4.2](#). $\square$

Corollary 5.3 (Exact optimality and phase saturation). *The component gap is zero if and only if*

$$\begin{aligned} C_s^* z_s &= 0, & \|z_s\|_{\infty} &\leq 1, \\ z_{s,i} &= y_{s,i}/|y_{s,i}| & &\text{whenever } y_{s,i} \neq 0. \end{aligned}$$

At coordinates where $y_{s,i} = 0$, only $|z_{s,i}| \leq 1$ is required. A zero global gap forces every component gap to vanish.

Proof. Equality in $\text{Re}(\bar{z}_i y_i) \leq |y_i|$ holds for nonzero y_i exactly at the displayed phase. Every component gap is nonnegative, so their sum vanishes exactly when each does. $\square$

5.1 What a numerical vector proves

A floating trial fill always gives a rigorous upper bound after its residual is evaluated with certified arithmetic. A floating charge with $C_s^* z_s \neq 0$, however small, is not a lower certificate for the unbounded parameter problem: the error term $[\theta_s, C_s^* z_s]_{\mathbb{R}}$ has no uniform bound. A lower certificate must lie exactly in $\text{Ker } C_s^*$, or be enclosed by a validated projection argument that proves exact dual feasibility.

Remark 5.4 (Solver status is not a theorem). Infeasibility, timeout, ill-conditioning, or failure to find a small upper bound supplies no obstruction. The obstruction is the explicit dual-feasible z_s and its checked objective.

6 Ambient tripartite graph versus actual incidence

Let K index the nonprimitive kernel sites. Write

$$B : \mathbb{C}^J \rightarrow \mathbb{C}^K, \quad A : \mathbb{C}^K \rightarrow \mathbb{C}^I, \quad C = AB.$$

For real-linear maps, $A_{ik} B_{kj}$ below means composition of the corresponding real coordinate blocks, and the sum is taken in $\text{Hom}_{\mathbb{R}}(\mathbb{C}, \mathbb{C})$. In the pair-dependent TPC-76 specialization all maps are complex-linear and these are the usual complex scalar entries.

Definition 6.1 (Ambient tripartite graph). The graph $\mathcal{G}_{A,B}$ has vertex set

$$J \sqcup K \sqcup I$$

and edges

$$j \sim k \iff B_{kj} \neq 0, \quad k \sim i \iff A_{ik} \neq 0.$$

Theorem 6.2 (Safe coarse factorization and refinement). *If $i \in I$ and $j \in J$ lie in different connected components of $\mathcal{G}_{A,B}$, then $C_{ij} = 0$. Consequently, the partition induced on $I \sqcup J$ by $\mathcal{G}_{A,B}$ gives a valid block decomposition of C . The actual graph $\mathcal{G}_C$ refines this partition and can split an ambient component further.*

Proof. Each summand in

$$C_{ij} = \sum_{k \in K} A_{ik} B_{kj}$$

can be nonzero only when there is a length-two path $j - k - i$ in $\mathcal{G}_{A,B}$. If i, j are in different components, every summand vanishes. Thus no actual edge crosses the ambient partition. The converse can fail because the nonzero summands can cancel, so $\mathcal{G}_C$ may contain fewer edges. $\square$

Example 6.3 (Exact cancellation splits an ambient block). *Take two sites, one analysis row, and one filter variable, with*

$$B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & -1 \end{pmatrix}.$$

The tripartite graph is connected, but $C = AB = 0$. The column is cost-null and the row is immutable. A graph built before multiplication misses this exact deletion.

Remark 6.4 (Unknown is not zero). When C_{ij} is known only inside an interval containing both zero and nonzero values, the edge must be retained. Deleting it by a tolerance can manufacture a false component factorization and a false dual certificate.

6.1 Assembly order

For the TPC application, the safe order is:

- (1) freeze the literal q -resolved carrier and raw normalization;
- (2) assemble B_L while retaining every declared (d, q) provenance column; exact complete-key cancellation may occur inside one fixed q -slice, but distinct q -columns are not merged;
- (3) assemble the full low-plaquette and exact-tail operator A_D ;
- (4) compute or validate $C = A_D B_L$;
- (5) delete exact zeros and build $\mathcal{G}_C$; and
- (6) solve and certify each resulting component.

Changing this order can confuse ambient support, actual support, and post- q aggregation.

7 Literal fixed-shift specialization and census protocol

7.1 Frozen physical data

Fix one nonzero h_0 , one declared external block, both orthogonal direct sides, and the complete TPC-74 missing-high keys. For every active unordered row pair $e = \{m, n\}$, freeze

$$\Delta_{mn} = h_0(n - m) \neq 0,$$

the literal slices $L_{mn,q}$, the rectangle B_{mn} , the cutoff D_{mn} , and all complete native key labels. Form

$$F_{mn}^0 \quad \text{from} \quad k_{mn} = \sum_{q \mid |\Delta_{mn}|} L_{mn,q},$$

and form $B_{L,mn}$ only from the actual directions $d \mid q \mid |\Delta_{mn}|$.

The normalization remains

$$\frac{1}{\sqrt{W_m W_n}},$$

where W_m, W_n are the global raw masses of all actual atoms on both direct sides. Analytic copies, graph vertices, and component counts do not enter these masses.

Corollary 7.1 (Literal edge decomposition). *With*

$$b_{mn} = A_{D_{mn}} F_{mn}^0, \quad C_{mn} = A_{D_{mn}} B_{L,mn},$$

the provenance gauge satisfies

$$\gamma_{mn}^{\text{prov}} = \|(b_{mn})_{\text{fix}}\|_1 + \sum_s \gamma_{mn,s},$$

and every $\gamma_{mn,s}$ has the exact matched certificate of [theorem 5.2](#).

Proof. All data are finite and have exactly the TPC-76 affine form. Apply [theorems 4.2](#) and [5.1](#). $\square$

7.2 Verified archive format

A finite census is reproducible only if it archives:

- (C1) the object identity: $h_0, m, n, \Delta_{mn}, B_{mn}, D_{mn}$, both-side block labels, complete-key identifiers, and raw W_m, W_n ;
- (C2) the literal q -slice table and exact or validated $F^0, B_L, A_D, C = A_D B_L$;
- (C3) the exact-zero policy, immutable-row ledger, zero-column ledger, and component index sets;
- (C4) for every component, $\theta_s, y_s, z_s, U_s, L_s$, together with checks of $y_s = b_s + C_s \theta_s$, $C_s^* z_s = 0$, $\|z_s\|_\infty \leq 1$, and the saturation defect $U_s - L_s$; phase saturation is required only when zero gap and optimality are claimed;
- (C5) the fixed-row cost and total certified gap;
- (C6) the Hermitian reverse checksum and assembled entry Q_{mn}^{prov} ; and
- (C7) an input-certification statement distinguishing literal coefficients from a rounded surrogate.

Remark 7.2 (No invented census). TPC-74 through TPC-76 provide symbolic exact formulas but no archived complete literal coefficient table for a growing physical carrier. Therefore the present paper proves the protocol and factorization theorems but reports no numerical physical component counts. A census of a toy or rounded instance would be labelled as such, not as the literal L1 carrier.

7.3 Matrix boundary

For independent unordered edges, the component sums assemble the entries of Q^{prov} , after which the TPC-76 Perron majorant applies. Components within one edge add to that edge cost. They do not imply

$$\rho\left(\sum_s Q^{(s)}\right) = \sum_s \rho(Q^{(s)}),$$

nor a direct-sum decomposition of the row matrix.

If a parameter is shared across edges, blocks, coefficient samples, or scales, one must first stack the entire coupled affine problem and build one global C . A shared column can connect several edge blocks. Pairwise components cannot be reused as a universal-filter theorem.

8 Countermodels and non-substitution laws

8.1 A connected ambient graph need not be actual

[Example 6.3](#) already shows that two nonzero paths can cancel in AB . Support multiplication in the Boolean semiring gives only an upper envelope for actual complex support.

8.2 A zero column need not be a zero fill direction

Let

$$A = \begin{pmatrix} 1 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The nonzero kernel direction $B\theta = (\theta, \theta)$ is annihilated by A . Deleting the column from the cost program is valid, but deleting it from an exported analytic completion changes the fill.

8.3 Small stationarity residual is not a lower bound

Take $C = [1]$, $b = 1$, and a real trial charge $z = \varepsilon$ with $0 < \varepsilon \leq 1$. Its apparent objective is ε , but $C^*z = \varepsilon \neq 0$. The true optimum is zero because $\theta = -1$. Indeed,

$$[z, b + C\theta]_{\mathbb{R}} = \operatorname{Re}(\varepsilon(1 + \theta))$$

is unbounded below along real θ when the erroneous stationarity term is separated. A tiny residual is not a tiny certificate error without a parameter bound.

8.4 Component factorization does not factor Perron radius

Let

$$Q^{(1)} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad Q^{(2)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Each has spectral radius 1, whereas

$$\rho(Q^{(1)} + Q^{(2)}) = \sqrt{2} \neq 2.$$

Adding certified edge-component costs is exact entrywise; adding their Perron radii is not.

8.5 Pairwise components can merge under a shared rule

Consider two scalar edge problems

$$|t|, \quad |1 - t|.$$

With independent variables each graph is a separate zero-cost component. A shared variable t connects the stacked rows into one component and the combined optimum is 1. Thus a pairwise incidence census cannot be promoted after the fact to a shared-filter result.

8.6 Finite sparsity is not asymptotic smallness

One immutable row with coefficient A_X forms a component of size one and contributes $|A_X|$. Its graph remains maximally sparse even when $|A_X|$ is comparable to the full raw mass. Component counts and normalized component cost are different statistics.

8.7 Rounded support can create false disconnection

For

$$C_\varepsilon = \begin{pmatrix} 1 & \varepsilon \\ 0 & 1 \end{pmatrix}, \quad \varepsilon \neq 0,$$

the actual graph is connected. Thresholding ε to zero splits it. Any certificate assembled from the false blocks is a theorem about the thresholded matrix, not about C_ε .

9 Claim ledger and next gate

9.1 Exact conclusions

The paper proves:

- (1) exact and distinct deletion rules for zero columns, immutable rows, and range-solvable blocks;
- (2) exact primal factorization over the connected components of the assembled nonzero graph of C ;
- (3) exact dual factorization over the same components, including the fixed-row phases;
- (4) componentwise and global matched fill/charge bounds with an additive certified gap;
- (5) phase saturation as the exact zero-gap condition;
- (6) a proof that the ambient tripartite graph is a safe coarse decomposition and that the actual C -graph refines it;
- (7) Hermitian transport of the component census;
- (8) an exact fixed-shift specialization to every finite literal TPC-76 provenance program; and
- (9) a reproducible archive protocol that distinguishes literal, validated, rounded, and toy inputs.

9.2 Statements not proved

The paper does not prove:

- a numerical census for a literal coefficient table that was not archived by the preceding papers;
- that most actual components are small or cheap;
- a mass-normalized component saving;
- decay of complete row sums or Perron radius;
- that edgewise components factor the row matrix;
- a single filter uniform over a growing coefficient class;
- that a failed solver run is a dual obstruction;
- that thresholded floating support equals literal support;
- closure of represented-frame, activation, missing-mass, nuisance, endpoint, or reconstruction gates;
- the strict endpoint budget below $1/400$;
- a statement special to $h_0 = 2$; or
- a parity-breaking, prime-pair, or twin-prime theorem.

9.3 Audit levels

Result	Scope	Level
Deletion and graph factorization	All finite affine ℓ^1 programs on complex coordinates with real-linear C	L0
Dual and gap certificates	All exact finite component blocks	L0
Ambient refinement theorem	All finite factorizations $C = AB$	L0
Literal fixed- h_0 protocol	Complete TPC-76 carrier when its coefficient table is supplied	L1
Actual growing-carrier census	Complete literal coefficient family	Open L1
Normalized/Perron saving	Complete growing row family	Open L2
Physical reassembly and endpoint	One common carrier and coefficient scope	Open

9.4 Next decision

The next useful question is structural rather than larger-scale numerics. Once a component is isolated, can its optimum or obstruction be read from a small exact motif? Candidate motifs include a surjective block, an immutable leaf, a one-dimensional quotient, a tree of incidences, and disjoint dual-charge packings. TPC-78 is planned to investigate exact motif reduction and to identify which local patterns can or cannot generate a nonzero cost.

9.5 Endpoint ledger

Component certification changes no downstream implication. Terminal activation, represented-frame recovery, missing-mass comparison, nuisance shorting, endpoint cells, and packet reconstruction remain open. If cumulative fixed-power loss reaches $1/400$, including equality, the current route stops.

9.6 Conclusion

The provenance completion gauge is no longer a black-box finite program. Its exact assembled incidence matrix

$$C = A_D B_L$$

determines a canonical coordinate graph, and every connected component carries its own attained primal and dual problem. Immutable costs, component optima, and certificate gaps add without approximation.

The distinction between the ambient tripartite graph and the actual product graph is essential. The former is safe for early coarse blocking; the latter is required for maximal exact deletion because complex paths can cancel. The distinction between a solver vector and a certificate is equally essential: only an exact annihilating charge provides a lower obstruction.

This advances the L1 route by making every future finite carrier census auditable. It does not supply the missing literal dataset, an asymptotic component law, or a physical saving. Those remain separate gates.

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